

Department of Information Systems

Business Data Telecommunication& Networks

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Network Performance Analysis

1. INSTALLATION OF WIRESHARK

Wireshark is a packet sniffer that is free for use and it is used to capture the Local Area Network (LAN) traffic. I have obtained it from official site at www.wireshark.org and then installed it on windows operating system. Wireshark helps to real-time capturing, filtering and analysis in detail of protocols this is why it is a key tool in measuring network performance.

2. QUALITY OF SERVICE (QOS) METRICS EXPLANATION

Quality of Service (QoS) parameters are vital parameters that measure and assess network performance:

Table 1: Quality of Service (QoS) metrics

Metric	Definition	Measurement Unit	Significance
Bandwidth	Maximum theoretical data transfer capacity of a network	Megabits per second (Mbps)	Indicates the upper limit of data that can be transmitted
Throughput	Actual achieved data transfer rate in real-world conditions	Megabits per second (Mbps)	Reflects real network performance after accounting for overhead, congestion, and errors
Latency	Time delay for a packet to travel from source to destination	Milliseconds (ms)	Affects real-time applications like video calls and online gaming
Packet Loss	Percentage of packets that fail to reach their destination	Percentage (%)	Causes retransmissions, reduced throughput, and poor user experience

3. SELECTED NETWORKS

Two networks were selected for performance comparison:

1. **Wi-Fi Network (Direct)** - Direct connection from computer to wireless router
2. **Mobile Hotspot** - Computer connected via smartphone hotspot (sharing the same Wi-Fi connection)

The mobile hotspot configuration adds an extra network hop (router → phone → computer), allowing analysis of how additional relay points affect performance.

Network Link: <https://www.wireshark.org/download.html>

4. WI-FI NETWORK ANALYSIS RESULTS

Packet Capture Screen

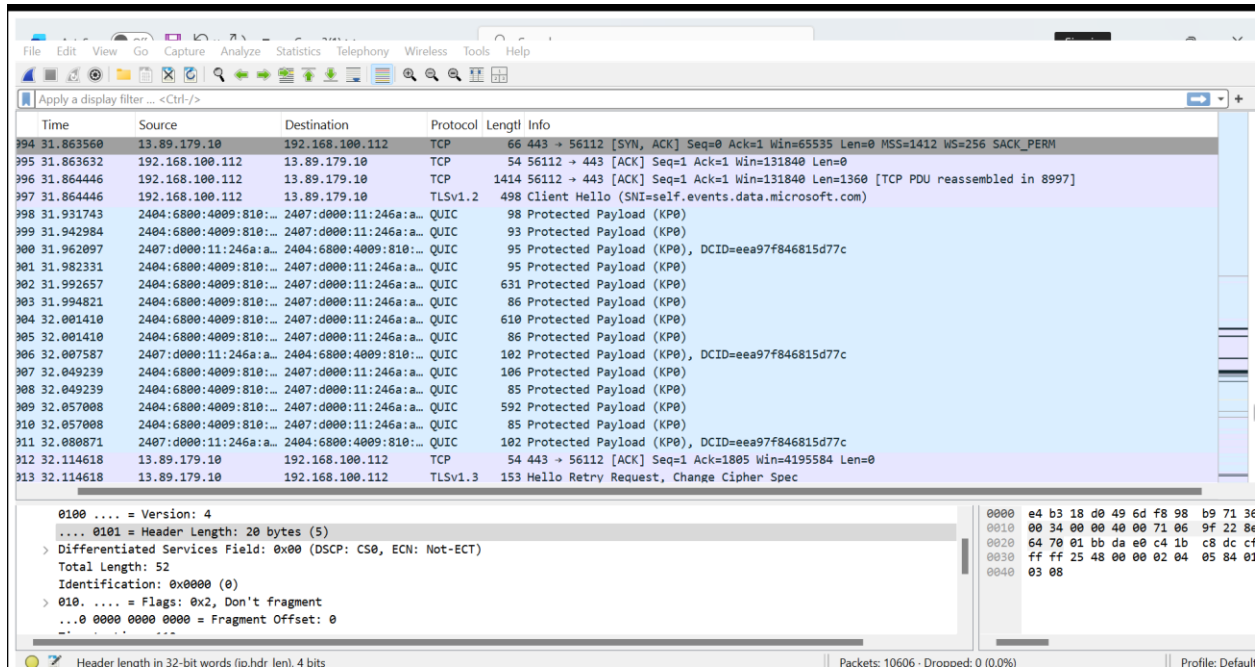


Figure 1: Wireshark Main Capture Window (Wi-Fi Network)

In this screenshot captured traffic is live on the Wi-Fi network with columns of packet number, time, source IP, destination IP, protocol, length and other information. There has been a mixture of IPv4 and IPv6 traffic in the capture and the protocols include SSDP, UDP, QUIC, and DNS.

Protocol Hierarchy Statistics

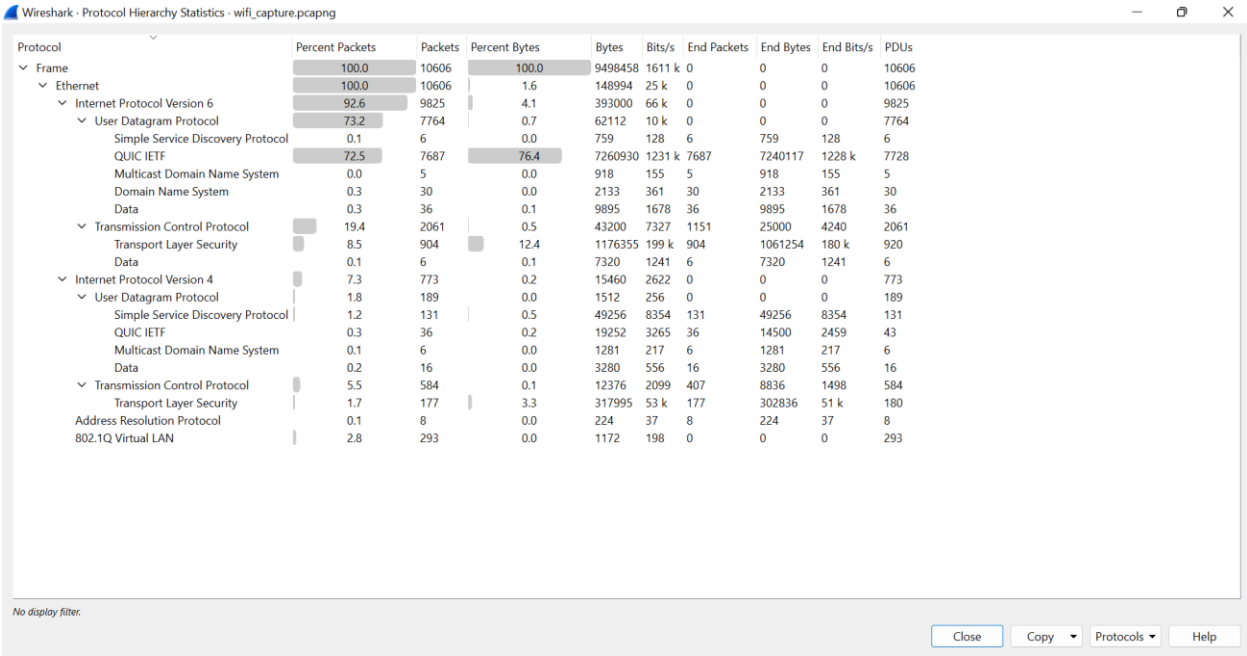


Figure 2: Protocol Hierarchy Statistics (Wi-Fi Network)

Table 2: Protocol Hierarchy Statistics - Wi-Fi Capture

Protocol	Percent Packets	Percent Bytes	Interpretation
Ethernet	100.0%	1.6%	All frames use Ethernet II framing
IPv6	92.6%	4.1%	Dominant by packet count but small packet sizes
UDP	73.2%	0.7%	Primary transport protocol for data transfer
QUIC IETF	72.5%	76.4%	Google's modern encrypted protocol
mDNS	0.1%	0.0%	Local network service discovery
TCP/TLS	5.5%	0.1%	Traditional encrypted web traffic

Analysis: Indicating heavy use of QUIC (HTTP/3) protocol, UDP leads at 73.2% of bytes. This up-to-date protocol decreases latency related to outdated TCP-based HTTPS.

Traffic Over Time (I/O Graph)

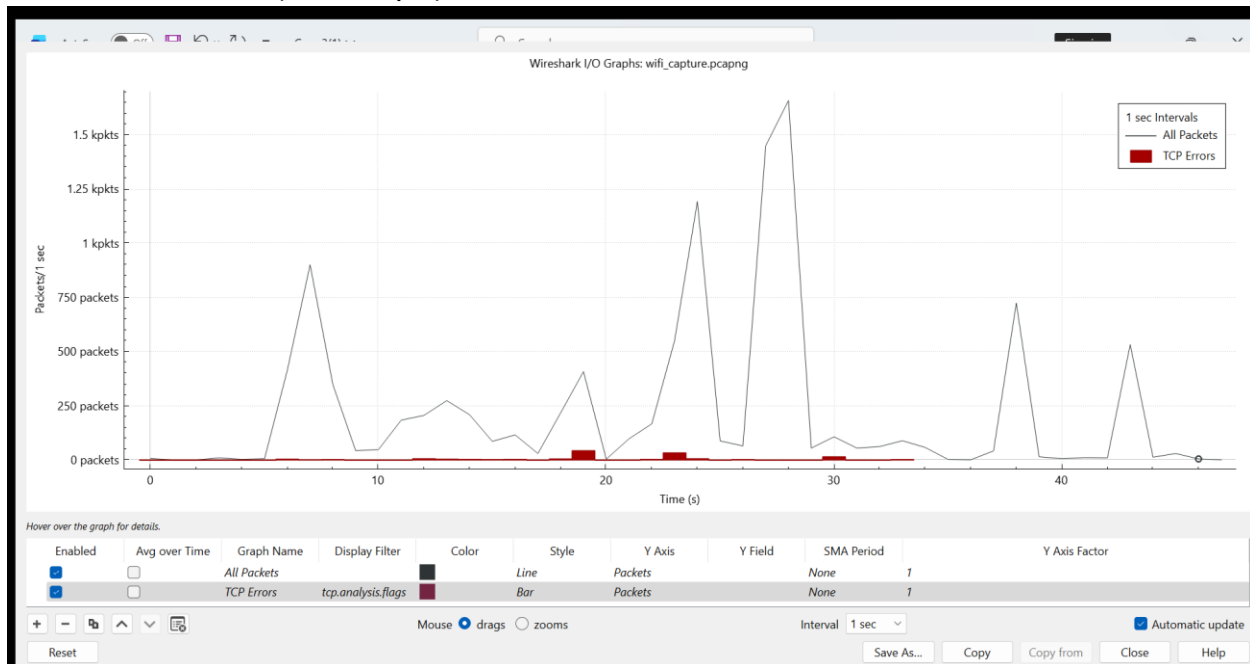


Figure 3: I/O Graph - Traffic Over Time (Wi-Fi Network)

The I/O Graph shows traffic in bits per second over the capture duration. Periodic bursts of activity correspond to web page loading events. The graph shows no TCP errors or retransmissions, indicating a stable connection.

Frame Details Analysis

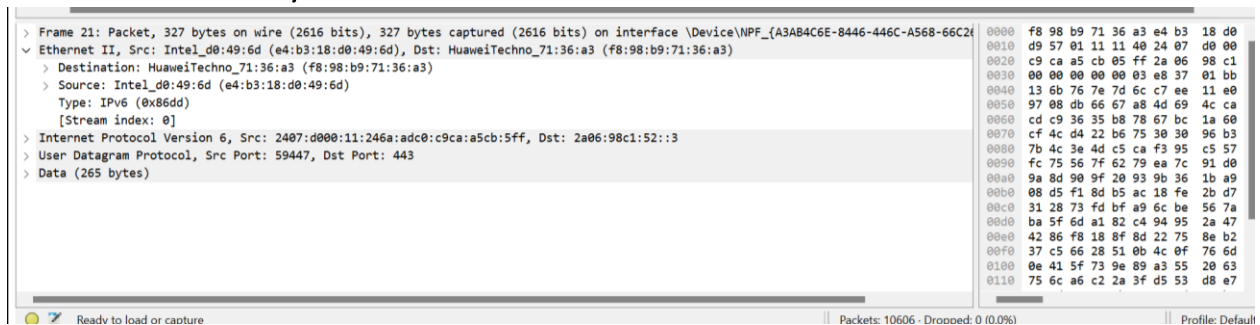


Figure 4: Frame Details (Wi-Fi Network)

The expanded frame section shows:

- **Frame size:** 327 bytes on wire, 327 bytes captured
- **Interface:** Wi-Fi network adapter
- **Frame metadata:** Includes arrival time and encapsulation type

Ethernet II Analysis

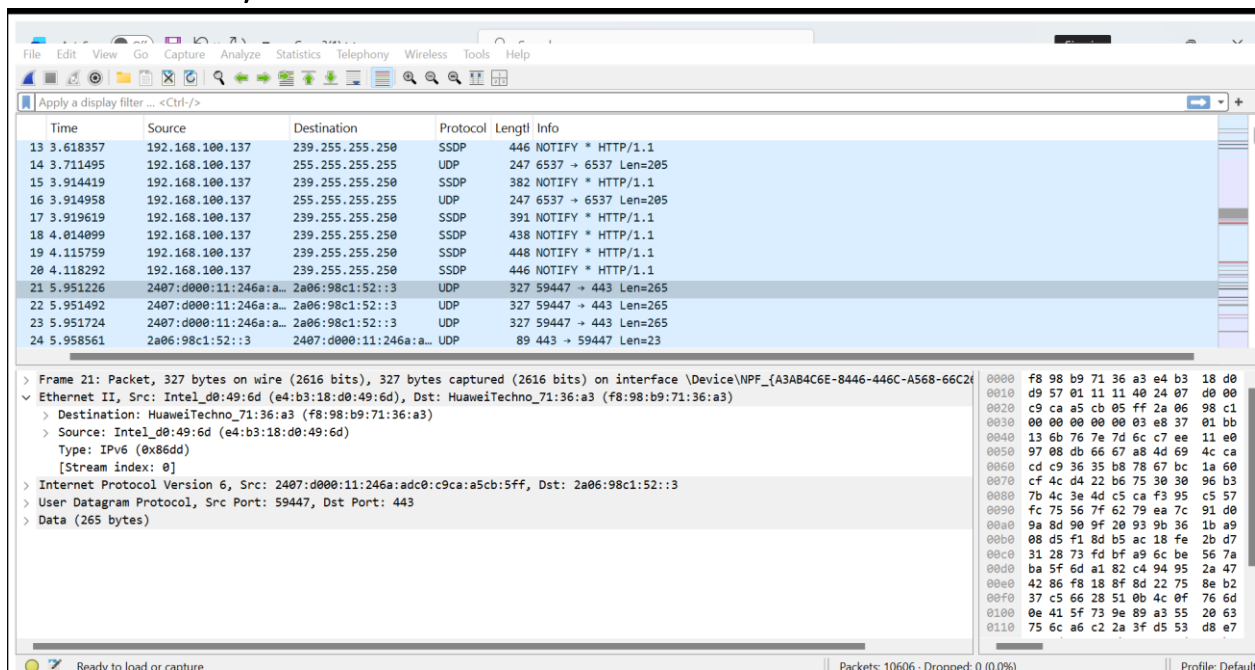


Figure 5: Ethernet II Header (Wi-Fi Network)

Table 3: Ethernet II Header (Wi-Fi Network)

Field	Value	Interpretation
Destination MAC	f8:98:b9:71:36:a3	device (Wi-Fi router)
Source MAC	e4:b3:18:d0:49:6d	Intel device (computer)
EtherType	0x86dd	IPv6 protocol

Analysis: The destination MAC address is the default gateway (router), confirming this packet is being sent from computer to router for internet forwarding.

Ping Test Results

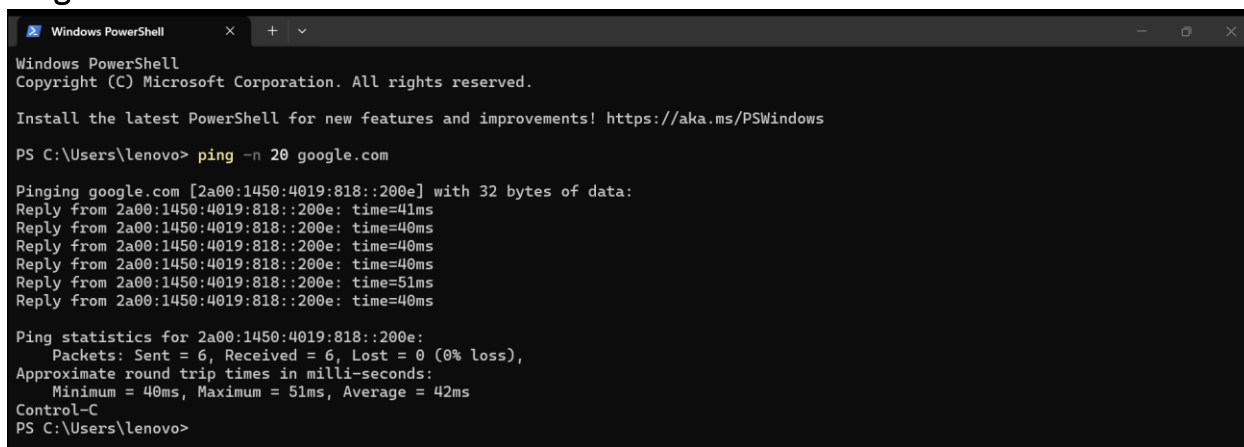


Figure 6: Ping Latency Test - Google Server (Wi-Fi Network)

Analysis: Zero packet loss indicates a steady connection with no overcrowding. Normal latency of 42ms is satisfactory for web browsing and streaming.

Traceroute Results

```
PS C:\Users\lenovo> tracert google.com

Tracing route to google.com [2a00:1450:4019:818::200e]
over a maximum of 30 hops:

  1     1 ms     1 ms     <1 ms   2407:d000:12:0:184b:b131:904d:58b8
  2     5 ms     9 ms     3 ms   2407:d000::172:16:86:57
  3     3 ms     2 ms     3 ms   2407:d000:1:1232::1
  4     3 ms    13 ms     3 ms   2407:d000:1:1231::2
  5     7 ms     3 ms     3 ms   2407:d000:0:160::1
  6     4 ms     4 ms     4 ms   2407:d000:0:9::1
  7    25 ms    26 ms    25 ms   2404:7000:f002:d3::1
  8    24 ms    24 ms    27 ms   2404:7000:f001:3::12:22
  9    27 ms    24 ms    25 ms   2404:7000:f001:1::4:18
 10   138 ms   138 ms   139 ms   2404:7000:f001:1::4:6
 11    40 ms    40 ms    40 ms   2001:4860:1:1::b22
 12    40 ms    40 ms    40 ms   2a00:1450:84c2:140::1
 13    40 ms    38 ms    39 ms   2001:4860:0:1::7da6
 14    41 ms    41 ms    49 ms   2001:4860:0:1::8374
 15    41 ms    41 ms    41 ms

PS C:\Users\lenovo> |
```

Figure 7: Traceroute (Wi-Fi Network)

The path to [google.com](https://www.google.com) shows 15 hops with latencies ranging from <1ms (local router) to 41ms (final destination). A spike to 138ms at hop 10 indicates international routing.

Speedtest Results

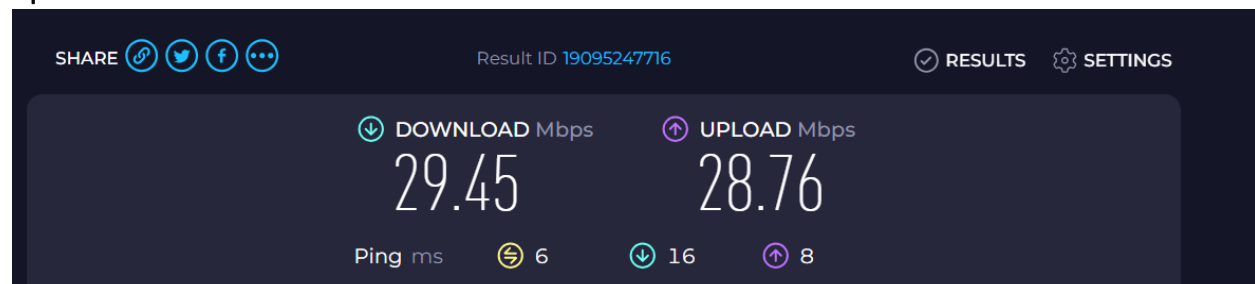


Figure 8: Speedtest.net Results (Wi-Fi Network)

Analysis: The resultant speed shows 29.45 download and 28.76 for uploading, that is perfect balanced speed for both tasks.

5. MOBILE HOTSPOT NETWORK ANALYSIS RESULTS

Packet Capture Screen

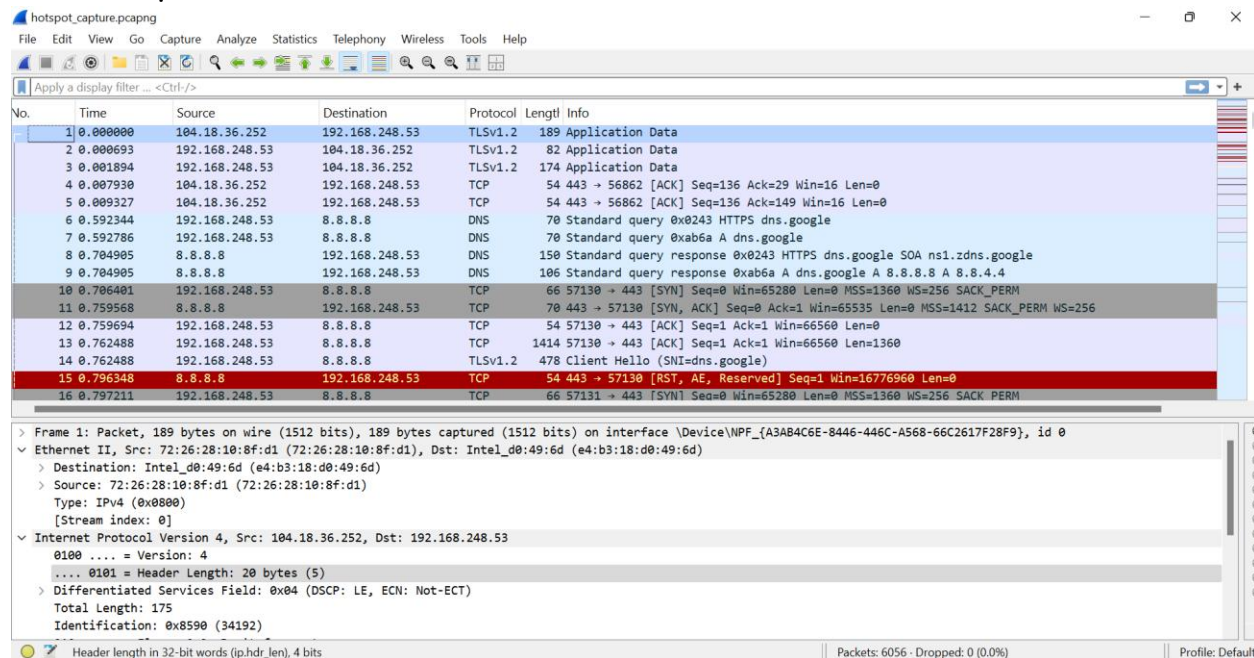


Figure 9: Packet Capture Screen (Mobile Hotspot)

This screen shot indicates the traffic that is captured by the mobile hot spot network. In above screenshot the origin IPv4 address of computer that is 192.168.248.53, while there were different destinations discovered like Google DNS 8.8.8.8 and Cloudflare 104.18.36.252.

Protocol Hierarchy Statistics

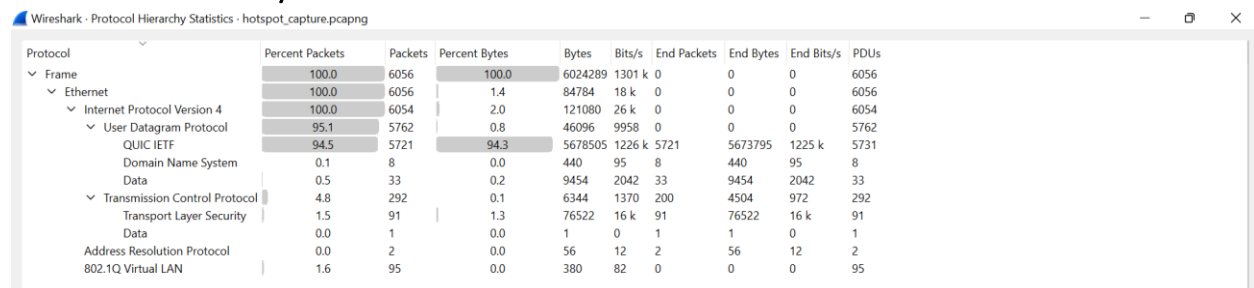


Figure 10: Protocol Hierarchy Statistics (Mobile Hotspot)

Table 4: Protocol Hierarchy Statistics (Mobile Hotspot)

Protocol	Percent Packets	Percent Bytes	Interpretation
Ethernet	100.0%	1.4%	All frames use Ethernet framing
IPv4	100.0%	2.0%	Only IPv4 traffic (no IPv6 on hotspot)
UDP	95.1%	0.8%	Dominant transport protocol

QUIC IETF	94.5%	94.3%	Primary data transfer protocol
TCP/TLS	4.8%	0.1%	Minor TCP traffic

Analysis: Similar to Wi-Fi, QUIC leads at 94.3% of bytes. Though, IPv6 is absent on the hotspot connection, indicating the hotspot relay does not pass IPv6 traffic.

Traffic Over Time (I/O Graph)

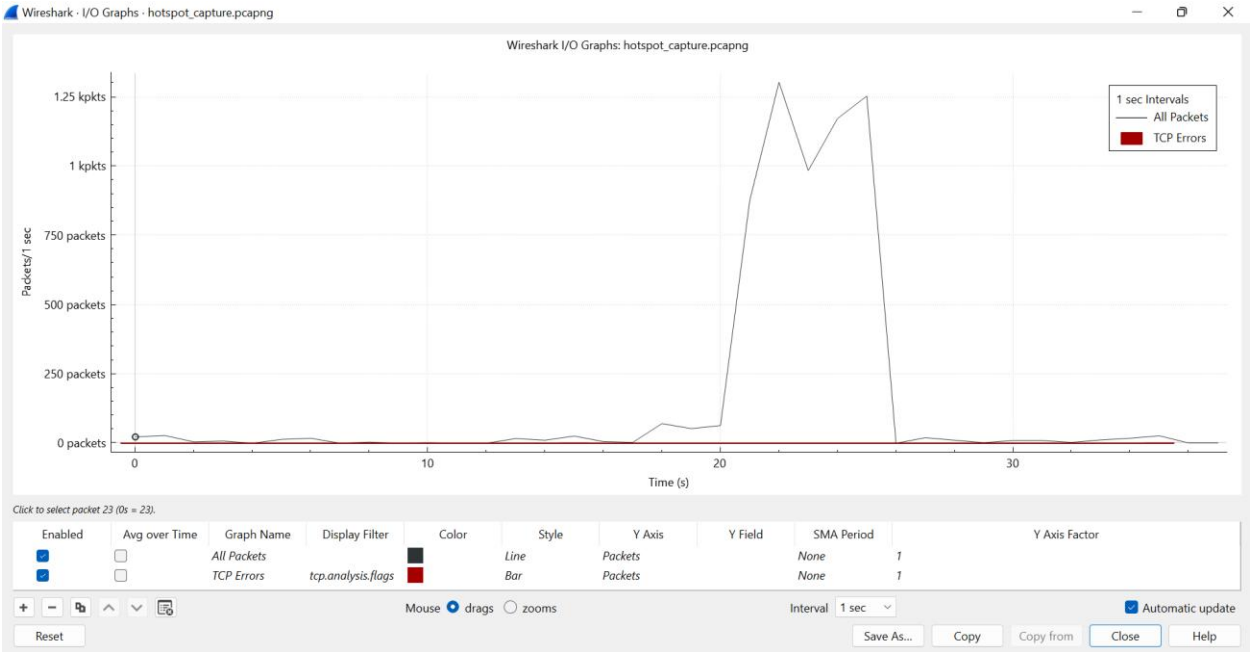


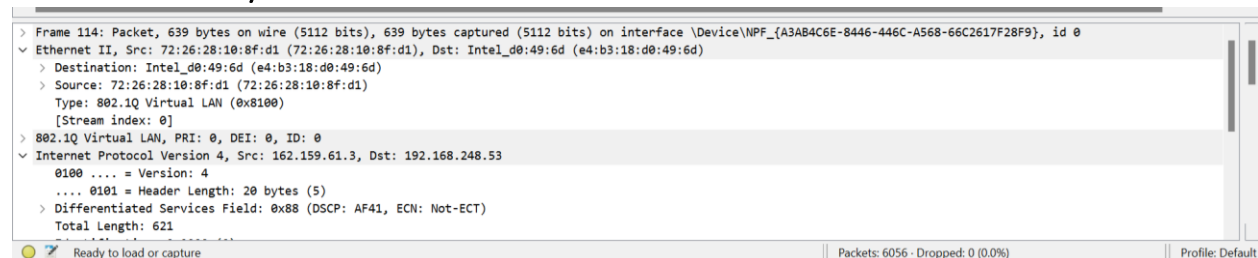
Figure 11: I/O Graph - Traffic Over Time (Mobile Hotspot)

The I/O Graph shows traffic patterns over the capture duration. The graph displays periodic traffic bursts similar to Wi-Fi but with lower peak amplitudes due to reduced throughput through the hotspot relay.

[illegible]

The expanded frame section shows:

- ## Ethernet II Analysis



Field	Value	Interpretation
Destination MAC	e4:b3:18:d0:49:6d	Intel device (computer)
Source MAC	72:26:28:10:8f:d1	Phone/ hotspot device
EtherType	0x8100	802.1Q Virtual LAN (VLAN tagging)

Analysis: The source MAC address is the phone sharing the hotspot. For traffic ranking, the EtherType 0x8100 specifies VLAN category.

Ping Test Results

```
Windows PowerShell
PS C:\Users\lenovo> ping -n 20 google.com

Pinging google.com [142.250.187.46] with 32 bytes of data:
Reply from 142.250.187.46: bytes=32 time=69ms TTL=108
Reply from 142.250.187.46: bytes=32 time=160ms TTL=108
Reply from 142.250.187.46: bytes=32 time=141ms TTL=108
Reply from 142.250.187.46: bytes=32 time=118ms TTL=108
Reply from 142.250.187.46: bytes=32 time=116ms TTL=108
Reply from 142.250.187.46: bytes=32 time=76ms TTL=108
Reply from 142.250.187.46: bytes=32 time=149ms TTL=108
Reply from 142.250.187.46: bytes=32 time=76ms TTL=108

Ping statistics for 142.250.187.46:
    Packets: Sent = 8, Received = 8, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 69ms, Maximum = 160ms, Average = 113ms
Control-C
PS C:\Users\lenovo> |
```

Figure 14: Ping Test (Mobile Hotspot)

Analysis: The zero packet loss percentage specifies that there is stable connectivity through the hotspot. However, regular latency increased from 42ms (Wi-Fi) to 113ms (hotspot), a 169% increase due to the extra relay hop.

Traceroute Results

```
PS C:\Users\lenovo> tracert google.com

Tracing route to google.com [142.250.187.46]
over a maximum of 30 hops:

  0  3 ms   2 ms   5 ms  192.168.248.97
  1  9 ms   3 ms   3 ms  192.168.100.1
  2 32 ms   5 ms   5 ms  100.95.32.1
  3  9 ms  10 ms   7 ms  172.31.8.2
  4  8 ms   9 ms   9 ms  172.31.8.9
  5 110 ms   8 ms  30 ms  172.31.5.254
  6  7 ms   7 ms   8 ms  172.31.5.170
  7 53 ms  77 ms  28 ms  10.95.178.65
  8 35 ms  60 ms  89 ms  10.253.4.26
  9 66 ms 126 ms  52 ms
10

PS C:\Users\lenovo> |
```

Figure 15: Traceroute (Mobile Hotspot)

In above screenshots the result of traceroute discovered that the Google.com route has 10 hops and the first hop is a phone/hotspot gateway with address 192.168.248.97. while their latencies were about 3ms-126ms. The traceroute has not gone through all hops which indicates that there could be packet filtering.

Speedtest Results

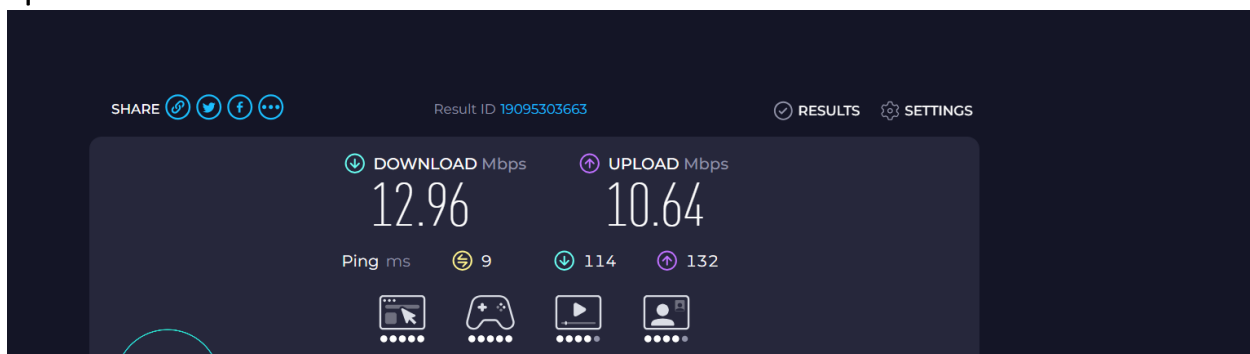


Figure 16: Speedtest (Mobile Hotspot)

Analysis: Throughput reduced considerably between 29.45 Mbps (Wi-Fi) and 12.96 Mbps (hotspot) by 56 per cent. Bufferbloat caused a tremendous increase in ping during active transfers.

6. PERFORMANCE METRICS COMPARISON

Table 5: Performance Metrics Comparison

Metric	Wi-Fi Network	Mobile Hotspot	Difference
Download Speed (Mbps)	29.45	12.96	-56%
Upload Speed (Mbps)	28.76	10.64	-63%
Average Ping to Google (ms)	42	113	+169%
Packet Loss (%)	0%	0%	No difference
Dominant Protocol	UDP/QUIC (76.4% bytes)	UDP/QUIC (94.3% bytes)	Similar
IPv6 Support	Yes	No	Hotspot lacks IPv6
Number of Hops	15	10	Hotspot has fewer visible hops

7. ANALYSIS AND DISCUSSION

Throughput Comparison

The Wi-Fi network achieved 29.45 Mbps download while the mobile hotspot achieved only 12.96 Mbps. This 56% reduction is caused by:

- **Double transmission:** Each packet is transmitted twice (router → phone → computer)
- **Medium contention:** Both hops share the same wireless spectrum
- **Processing delay:** The phone must process and forward each packet

Latency Comparison

Average latency increased from 42ms (Wi-Fi) to 113ms (hotspot). The additional delay comes from:

- **Processing time** at the phone (5-15ms per packet)
- **Queuing delay** (bufferbloat) in the phone's network stack
- **Contention** between the two wireless links

Protocol Analysis

In both networks QUIC (UDP-based) is the most dominant protocol. The hot spot is however not IPv6 compliant and all traffic must be IPv4 with extra encapsulation overhead.

8. RECOMMENDATIONS TO IMPROVE NETWORK PERFORMANCE

According to the analysis the recommendations to the Mobile Hotspot network (the weaker performer) are as follows:

USB Tethering, rather than Wi-Fi Hotspot.	Problem at hand: Wi-Fi hotspot needs two-wireless transmissions.
	Solution: Use USB cable and turn on USB tethering.
	Improvement to be expected: 30-50 percent decrease in latency, 20-30 percent improvement in throughput.
Minimise Physical Distance.	Problem at hand: Signal degradation is an issue to throughput.
	Solution: Have phone within 1-2 feet of the computer.
	Expected improvement: 10-20 percent increase in throughput.
Minimize Interference	Current problem: There is contention with other Wi-Fi networks.
	Solution: 5GHz band is preferred to 2.4GHz, when possible.
	Predicted enhancement: Less latency and an augmented throughput.
Turn off Apps in the Background on phone.	Problem at hand: Phone is used to handle other traffic.
	Resolution: Turn all the background applications off when using the hotspot.
	Latency improvement to be expected: 10-15% reduction.
Turn off IPv6 on Computer.	Problem at hand: Hotspot does not support IPv6 and this results in fallback delays.
	Solution: Turn off IPv6 on network adapter.
	Improvement anticipated: There should be less time required to set up connections.

Figure 17: RECOMMENDATIONS TO IMPROVE NETWORK PERFORMANCE

9. CONCLUSION

This network performance test contrasted direct Wi-Fi connection with mobile hotspot (Wi-Fi sharing) via Wireshark packet capture, ping, traceroute and Speedtest applications.

Key Findings:

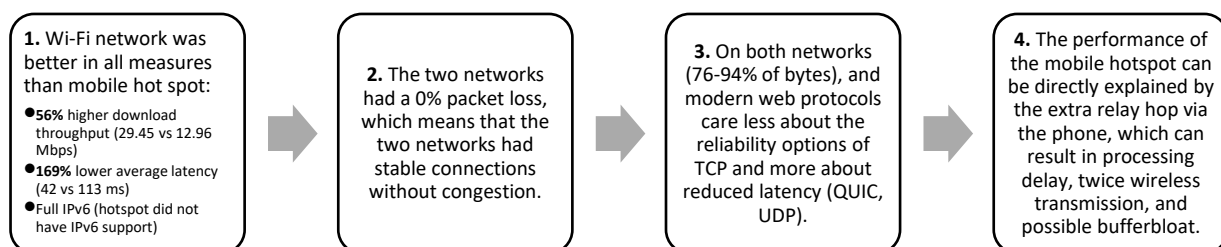


Figure 18: Key Findings of Overall Report

Final Conclusion: To perform optimally, it is suggested to use direct Wi-Fi connection. Mobile hotspots are not recommended when Wi-Fi is available but should be used only when there is no Wi-Fi and should be able to implement the above recommendations to reduce losses in performance.